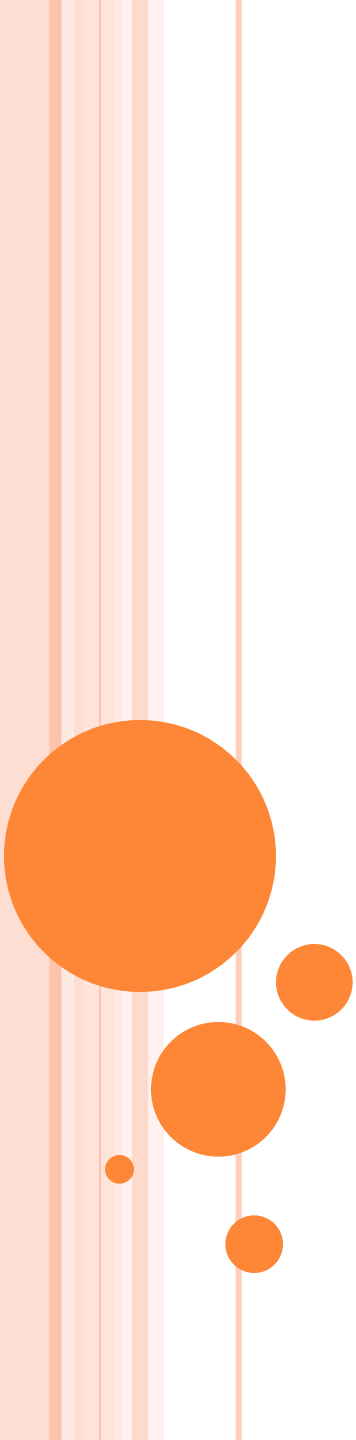


A decorative graphic on the left side of the slide consisting of several vertical orange lines of varying heights and a cluster of five orange circles of different sizes.

FIRST-ORDER (PREDICATE) LOGIC

TRANSFORMATION OF NATURAL LANGUAGE SENTENCES INTO PREDICATE FORMULAS

1. **All** Computer Science (CS) students are smart.
2. **There is** someone who studies at Babes-Bolyai University (*BBU*) and is smart.
3. If x and y are **nonnegative integers** and x is **greater than** y , then x^2 is **greater than** y^2 .
4. In a plane if a line x is **parallel** to a constant line d then **all** the lines **perpendicular** to x are also **perpendicular** to d .
5. **Every** child loves **anyone** who gives the child **any** present.



THE AXIOMS THAT DEFINE THE NATURAL NUMBERS

a₁. Every natural number has a unique immediate successor.

existence: $(\forall x)(\exists y)equal(y, successor(x))$

unicity: $(\forall x)(\forall y)(\forall z)(equal(y, successor(x)) \wedge equal(z, successor(x)) \rightarrow equal(y, z))$

a₂. The number 0 is *not* the immediate successor of a natural number.

$\neg(\exists x)equal(0, successor(x))$

a₃. Every natural number, except 0, has a unique immediate predecessor.

existence: $(\forall x)(\exists y)(\neg equal(0, x) \wedge equal(y, predecessor(x)))$

unicity: $(\forall x)(\forall y)(\forall z)(equal(y, predecessor(x)) \wedge equal(z, predecessor(x)) \rightarrow equal(y, z))$

unary functions: *successor, predecessor* ;

binary predicate: *equal*.

The predicate *equal* is defined by the following axioms:

$(\forall x)equal(x, x)$ – **reflexivity**

$(\forall x)(\forall y)(equal(x, y) \rightarrow equal(y, x))$ – **symmetry**

$(\forall x)(\forall y)(\forall z)(equal(x, y) \wedge equal(y, z) \rightarrow equal(x, z))$ – **transitivity**

The equality of the successors and the predecessors of two equal numbers:

$(\forall x)(\forall y)(equal(x, y) \rightarrow equal(successor(x), successor(y)))$

$(\forall x)(\forall y)(equal(x, y) \rightarrow equal(predecessor(x), predecessor(y)))$



AXIOMATIC SYSTEM OF PREDICATE LOGIC $\text{Pr} = (\Sigma_{\text{Pr}}, F_{\text{Pr}}, A_{\text{Pr}}, R_{\text{Pr}})$

$\Sigma_{\text{Pr}} = \text{Var} \cup \text{Const} \cup (\bigcup_{j=1}^n F_j) \cup (\bigcup_{j=1}^m P_j) \cup \text{Connectives} \cup \text{Quantifiers}$ - **vocabulary**

- **Var** is the set of **variable** symbols $\{x, y, z, \dots\}$;
- **Const** is the set of **constants** $\{a, b, c, \dots\}$;
- $F_i = \{f \mid f : D^i \rightarrow D\}$ is the set of **function symbols** of arity “ i ”
- $P_i = \{p \mid p : D^i \rightarrow \{T, F\}\}$ is the set of **predicate symbols** of arity “ i ”
- **Connectives** = $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$;
- **Quantifiers** = $\{\forall(\text{universal quantifier}), \exists(\text{existential quantifier})\}$

The priority of the quantifiers is greater than the priorities of the connectives.

○ **TERMS** is the set of terms:

- $\text{Var} \subset \text{TERMS}$; $\text{Const} \subset \text{TERMS}$;
 - if $f \in F_k$ and $t_1, t_2, \dots, t_k \in \text{TERMS}$ then $f(t_1, t_2, \dots, t_k) \in \text{TERMS}$
- examples:** $x, a, f(x), g(x, a), g(f(x), y)$

○ **ATOMS** is the set of atomic formulas (atoms):

- $T, F \in \text{ATOMS}$
 - if $P \in P_k$ and $t_1, \dots, t_k \in \text{TERMS}$ then $P(t_1, \dots, t_k) \in \text{ATOMS}$
- examples:** $T, F, P(x, y, a), Q(f(x), a), R(a, g(f(x), y))$

○ **Literal** - an atom or its negation.

examples: $P(f(x), a, y, g(x, b)), \neg Q(x, a, g(x))$



AXIOMATIC SYSTEM (CONTD.) $\text{Pr} = (\Sigma_{\text{Pr}}, F_{\text{Pr}}, A_{\text{Pr}}, R_{\text{Pr}})$

- F_{Pr} is the *set of well-formed formulas*
- $A_{\text{Pr}} = \{A_1, A_2, A_3, A_4, A_5\}$ is the *set of axioms*
 - $A_1 : U \rightarrow (V \rightarrow U)$
 - $A_2 : ((U \rightarrow (V \rightarrow Z)) \rightarrow ((U \rightarrow V) \rightarrow (U \rightarrow Z)))$
 - $A_3 : (U \rightarrow V) \rightarrow (\neg V \rightarrow \neg U)$ (*modus tollens*)
 - $A_4 : (\forall x)U(x) \rightarrow U(t)$, where t is a term.
(*universal instantiation*)
 - $A_5 : (U \rightarrow V(y)) \rightarrow (U \rightarrow (\forall x)V(x))$, where x is not free in U or V ,
 y is free in V and does not appear in U .
- $R_{\text{Pr}} = \{mp, gen\}$ is the *set of inference rules*:
 - *modus ponens* symbolized as: $U, U \rightarrow V \vdash_{mp} V$
 - *universal generalization rule* symbolized as:
 $U(x) \vdash_{gen} (\forall x)U(x)$, x is a free variable in U



OPEN VERSUS CLOSED FORMULAS

Definition:

1. In a predicate formula the variables which are within the scope of a quantifier are called *bound variables*, all the others are called *free variables*.
2. A formula is called a *closed formula* if all its variables are bound.
3. If a formula contains at least one free variable, the *formula is open*.

Example:

1. The predicate formula $(\forall x)(\exists z)(P(x, z, a) \vee (\exists y)Q(x, f(y)))$ is *closed*
 - where: $x, y, z \in Var, a \in Const, f \in F_1, P \in P_3, Q \in P_2$.
 - all variables (x, y, z) are bound.
2. The predicate formula $(\forall x)P(x, y) \wedge Q(z, a)$ is *open*,
 - where: $a \in Const, x, y, z \in Var, P, Q \in P_2$
 - the variables y and z are *free*,
 - x is a *bound variable* (within the scope of $\forall$)



DEDUCTION IN FIRST-ORDER LOGIC

Definition:

Let $U_1, U_2, \dots, U_n, V$ be first-order formulas, $U_1, U_2, \dots, U_n$ are the *hypotheses (premises)* and V is the *conclusion*. V is *deducible (inferable, derivable)* from $U_1, U_2, \dots, U_n$: notation: $U_1, U_2, \dots, U_n \vdash V$, if there exists a sequence of formulas $(f_1, f_2, \dots, f_m)$ such that $f_m = V$ and $\forall i \in \{1, \dots, m\}$ we have a) or b) or c) or d).

- a) $f_i \in A_{Pr}$ (axiom of predicate logic);
- b) $f_i \in \{U_1, U_2, \dots, U_n\}$ (hypothesis formula);
- c) $f_{i_1}, f_{i_2} \vdash_{mp} f_i$, $i_1 < i$ and $i_2 < i$ (formula f_i is inferred, using *modus ponens* rule, from two formulas that exist in the sequence);
- d) $f_j \vdash_{gen} f_i$, $j < i$ (formula f_i is obtained using the *universal generalization* rule from a formula that exists already in the sequence).

The sequence $(f_1, f_2, \dots, f_m)$ is called the *deduction* of V from $U_1, U_2, \dots, U_n$.

Definition:

A formula $U \in F_{Pr}$, such that $\emptyset \vdash U$ (notation: $\vdash U$) is called a *theorem*.

Remark: The theorems are the formulas deducible from the axioms, using *modus ponens* and the *generalization* rule.

INFERENCE RULES IN FIRST-ORDER LOGIC

	Inference rule
<i>universal instantiation</i>	$(\forall x)U(x) \vdash_{univ_inst} U(t) ,$ <i>t</i> is a <i>term</i> (variable or constant of the domain)
<i>universal generalization</i>	$U(x) \vdash_{univ_gen} (\forall x)U(x) ,$ <i>x</i> is a <i>free variable</i> in <i>U</i>
<i>existential instantiation</i>	$(\exists x)U(x) \vdash_{exist_inst} U(c) ,$ <i>c</i> is a <i>new constant</i> of the domain
<i>existential generalization</i>	$U(t) \vdash_{exist_gen} (\exists x)U(x) ,$ <i>t</i> is a <i>variable</i> or a <i>constant</i> of the domain, the variable <i>x</i> must not appear free in <i>U</i>



EXAMPLE 1 – MODELING REASONING

Hypotheses:

H_1 . All hummingbirds are richly colored.

H_2 . No large birds live on honey.

H_3 . Birds that do not live on honey are dull in color.

H_4 . *Piky* is a hummingbird.

Conclusions:

C_1 . There is a bird which lives on honey.

C_2 . All hummingbirds are small.

Check whether the following deductions hold or not.

$H_1, H_2, H_3, H_4 \vdash C_1$ and $H_1, H_2, H_3 \vdash C_2$.



HUMMING BIRDS – AMAZING FACTS

<https://www.youtube.com/watch?v=FPRswRWZ23Q>

- The name, **hummingbird**, comes from the humming noise their **wings** make as they beat so fast, **60-80 beats per second**.
- They are **the smallest migrating birds in the world**. The average weight of a hummingbird is 3 grams, less than a nickel.
- Unlike other birds, hummers can rotate their wings 180 degrees. This means that they **can fly forward and backward, and also side to side**. Plus, they can easily hover in the same spot when they're in front of flowers and feeding on nectar.
- They **have no sense of smell**, but they have **extremely keen eyesight**.
- Hummingbirds **have the largest heart of any bird for its size**. Its heart rate is often more than 1,200 beats per minute.
- They are **very intelligent**, and are able to remember places and individual people from one year to the next. This is because a large portion of a hummingbird's brain is occupied by the hippocampus, an area dedicated to learning and spatial memory.
- A hummingbird's daytime **body temperature** ranges from 105 to 108 F.
- There are **340 species** of hummingbirds making them the second most diverse bird family on earth.



EXAMPLE 1 –MODELING REASONING (contd.)

Hypotheses:

- H_1 . All hummingbirds are richly colored.
 H_2 . No large birds live on honey.
 H_3 . Birds that do not live on honey are dull in color.
 H_4 . *Piky* is a hummingbird.

Conclusions:

- C_1 . There is a bird which lives on honey.
 C_2 . All hummingbirds are small.
 Check whether the following deductions hold or not.

$$H_1, H_2, H_3, H_4 \vdash C_1 \text{ and } H_1, H_2, H_3 \vdash C_2.$$

First-order (predicate) formulas:

- $H_1 : (\forall x)(hb(x) \rightarrow rc(x))$
 $H_2 : \neg(\exists x)(\neg sb(x) \wedge lh(x)) \equiv$
 $\equiv (\forall x)(\neg sb(x) \rightarrow \neg lh(x))$
 $H_3 : (\forall x)(\neg lh(x) \rightarrow \neg rc(x))$
 $H_4 : hb(Piky)$
 $C_1 : (\exists x)lh(x)$
 $C_2 : (\forall x)(hb(x) \rightarrow sb(x))$

D is the **domain** (the universe of birds)

Piky is a **constant** of the universe.

unary **predicate symbols**: *hb*, *rc*, *sb*, *lh*.

$hb : D \rightarrow \{T, F\}$, $hb(x) = T$ if x is a *hummingbird*,

$rc : D \rightarrow \{T, F\}$, $rc(x) = T$ if x is *richly colored*,

$sb : D \rightarrow \{T, F\}$, $sb(x) = T$ if x is a *small bird*,

$lh : D \rightarrow \{T, F\}$, $lh(x) = T$ if x *lives on honey*.

EXAMPLE 1 – BUILDING DEDUCTIONS (LECTURE)

Inference rules
$(\forall x)U(x) \vdash_{univ_inst} U(t)$, <i>t</i> is a term (variable or constant of the domain)
$U(x) \vdash_{univ_gen} (\forall x)U(x)$, <i>x</i> is a free variable in <i>U</i>
$(\exists x)U(x) \vdash_{exist_inst} U(c)$, <i>c</i> is a new constant of the domain
$U(t) \vdash_{exist_gen} (\exists x)U(x)$, <i>t</i> is a variable or a constant of the domain
$U, U \rightarrow V \vdash_{mp} V$ (modus ponens)
$U \rightarrow V, V \rightarrow Z \vdash_{syllogism} U \rightarrow Z$
$U \rightarrow V \vdash_{mt} \neg V \rightarrow \neg U$ (modus tollens) $\neg V, U \rightarrow V \vdash_{mt} \neg U$

First-order (predicate) formulas:

$$H_1 : (\forall x)(hb(x) \rightarrow rc(x))$$

$$H_2 : \neg(\exists x)(\neg sb(x) \wedge lh(x)) \equiv (\forall x)(\neg sb(x) \rightarrow \neg lh(x))$$

$$H_3 : (\forall x)(\neg lh(x) \rightarrow \neg rc(x))$$

$$H_4 : hb(Piky)$$

$$C_1 : (\exists x)lh(x)$$

$$C_2 : (\forall x)(hb(x) \rightarrow sb(x))$$



EXAMPLE 2



M. Lupea

Using the definition of deduction and the inference rules prove that:

$$(\forall x)(\forall y)(P(x) \vee Q(y)), \neg(\forall z)P(z) \vdash (\exists t)Q(t)$$

The sequence $(f_1, f_2, f_3, f_4, f_5, f_6, f_7)$ of predicate formulas is generated.

$$f_1 : (\forall x)(\forall y)(P(x) \vee Q(y)) \text{ -- hypothesis}$$

$$f_2 : \neg(\forall z)P(z) \equiv (\exists z)\neg P(z) \text{ -- hypothesis}$$

We have a universal formula (f_1) and an existential formula (f_2). It is recommended to use first the *existential instantiation* inference rule that introduces a new constant, and then this constant will be used for *universal instantiations*.

$$f_2 \vdash_{\text{exist_inst}} f_3 = \neg P(c), \text{ } c \text{ is a new constant;}$$

$$f_1 \vdash_{\text{univ_inst}} f_4 = (\forall y)(P(c) \vee Q(y)),$$

the outermost universally quantified variable x was instantiated using the constant c ;

$$f_4 \vdash_{\text{univ_inst}} f_5 = P(c) \vee Q(c),$$

the universally quantified variable y was instantiated using the constant c ;

$$f_5 \equiv \neg P(c) \rightarrow Q(c)$$

$$f_3, f_5 \vdash_{\text{mp}} f_6 = Q(c), \text{ } \textit{modus ponens} \text{ was applied;}$$

$$f_6 \vdash_{\text{exist_gen}} f_7 = (\exists t)Q(t),$$

existential generalization was applied, t is a variable introduced by this inference rule;

$(f_1, f_2, f_3, f_4, f_5, f_6, f_7)$ is the deduction (proof) of the conclusion f_7 from the hypotheses f_1 and f_2 .

EXAMPLE 3: SUCCESSION TO THE BRITISH THRONE

Hypotheses:

H1: If x is the king and y is his oldest son, then y can become the king.

H2: If x is the king and y defeats x , then y will become the king

H3: *Richard III* is the king.

H4: *Henry VII* defeated *Richard III*.

H5: *Henry VIII* is *Henry VII*'s oldest son.

Conclusion C: Can *Henry VIII* become the king?



KINGS OF ENGLAND

Richard III (1452 – 1485)

- He was the last king of the [House of York](#) and the last of the [Plantagenet dynasty](#).
- His defeat and death at the [Battle of Bosworth Field](#), the last decisive battle of the [Wars of the Roses](#), marked the end of the Middle Age in England.
- He is the protagonist of [Richard III](#), one of [William Shakespeare's](#) history plays.

Henry VII (1457 – 1509)

- He defeated Richard III and was the last king of England to win his throne on the field of battle.
- He was the first monarch of the [House of Tudor](#).
- Henry VII's policy was both to maintain peace and to create economic prosperity.

Henry VIII (1491 – 1547)

- He is best known for his six marriages.
- He initiated the [English Reformation](#), separating the [Church of England](#) from papal authority.
- Domestically, Henry is known for his radical changes to the [English Constitution](#), ushering in the theory of the [divine right of kings](#).
- He has been described as "[one of the most charismatic rulers to sit on the English throne](#)" and his reign has been described as the "most important" in English history. He was an author and composer.
- [Thomas More](#) was one of his chief ministers, executed after refusing to take the [Oath of Supremacy](#). [Thomas More](#) was an English lawyer, judge, social philosopher, author, statesman. He wrote [Utopia](#), published in 1516, which describes the political system of an imaginary island state.



THEOREMS

Theorem of deduction:

If $U_1, \dots, U_{n-1}, U_n \vdash V$, then $U_1, \dots, U_{n-1} \vdash U_n \rightarrow V$.

Reverse of the theorem of deduction:

If $U_1, \dots, U_{n-1} \vdash U_n \rightarrow V$ then $U_1, \dots, U_{n-1}, U_n \vdash V$.

Refutation theorem:

If $U_1, \dots, U_{n-1}, U_n \cup \{\neg V\}$ is inconsistent then $U_1, \dots, U_{n-1}, U_n \vdash V$.

Note: The last theorem models “*reductio ad absurdum principle*”
(proof by contradiction) and is used in proof methods such as:

- *resolution*,
 - *semantic tableaux method*,
- called *refutation proof methods*.



Example 4: Using the theorem of deduction, prove that the formula
 $(\forall x)(A(x) \rightarrow B(x)) \rightarrow ((\forall x)A(x) \rightarrow (\forall x)B(x))$ is a theorem.

Step1:

We apply the reverse of the theorem of deduction:

if $\vdash (\forall x)(A(x) \rightarrow B(x)) \rightarrow ((\forall x)A(x) \rightarrow (\forall x)B(x))$

then $(\forall x)(A(x) \rightarrow B(x)) \vdash (\forall x)A(x) \rightarrow (\forall x)B(x)$

then $(\forall x)(A(x) \rightarrow B(x)), (\forall x)A(x) \vdash (\forall x)B(x) (*)$

Step3:

Using the deduction (*) proved at Step 2 we apply twice the theorem of deduction:

if $(\forall x)(A(x) \rightarrow B(x)), (\forall x)A(x) \vdash (\forall x)B(x)$ then

$(\forall x)(A(x) \rightarrow B(x)) \vdash (\forall x)A(x) \rightarrow (\forall x)B(x)$ then

$\vdash (\forall x)(A(x) \rightarrow B(x)) \rightarrow ((\forall x)A(x) \rightarrow (\forall x)B(x))$

We have proved that the initial formula is a theorem.

Step2: We prove

$(\forall x)(A(x) \rightarrow B(x)), (\forall x)A(x) \vdash (\forall x)B(x) (*)$

using the definition of a deduction.

The sequence $(f1, f2, \dots, f8)$ is generated:

f1: $(\forall x)(A(x) \rightarrow B(x))$ - hypothesis

f2: $(\forall x)(A(x) \rightarrow B(x)) \rightarrow (A(y) \rightarrow B(y))$

-- axiom A4, x is instantiated with y

f1, f2 $\vdash_{mp}$ f3 = $A(y) \rightarrow B(y)$

f4: $(\forall x)A(x)$ - hypothesis

f5: $(\forall x)A(x) \rightarrow A(y)$ --- axiom A4

f4, f5 $\vdash_{mp}$ f6 = $A(y)$

f3, f6 $\vdash_{mp}$ f7 = $B(y)$

f7 $\vdash_{gen}$ f8 = $(\forall x)B(x)$

THE SEMANTICS OF FIRST-ORDER LOGIC

- The semantics of predicate logic realize the connection between the *constant symbols*, the *function symbols*, the *predicate symbols* and the *real constants*, *functions*, *predicates* from the modeled universe.
- It is provided a meaning in terms of the modeled universe for each formula from the language.

Definition:

An *interpretation* of predicate formula is a pair $I = \langle D, m \rangle$, where:

- D is a nonempty set called the domain of interpretation.
- m is a function that assigns:
 - a fixed value $m(c) \in D$ to the constant c .
 - a function $m(f): D^n \rightarrow D$ to each n -ary function symbol f ;
 - a predicate $m(P): D^n \rightarrow \{T, F\}$ to each n -ary predicate symbol P .



Notations: Let $I = \langle D, m \rangle$ be an interpretation.

- $|I| = D$ is the domain of I , Var is the set of variables.
- $I|X|$ is $m(X)$ where X is a predicate symbol or a function symbol.
- $As(I)$ is the set of assignment functions for variables over the domain of I .
 $a \in As(I), a: Var \rightarrow |I|$.
- $[a]_x = \{a' \mid a' \in As(I) \text{ and } a'(y) = a(y), \text{ for every } y \neq x\}$.

Definition: Let I be an interpretation and $a \in As(I)$. The evaluation function v_a^I is defined:

- $v_a^I(x) = a(x), x \in Var$; $v_a^I(c) = I|c|, c \in Const$;
- $v_a^I(f(t_1, t_2, \dots, t_n)) = I|f|(v_a^I(t_1), v_a^I(t_2), \dots, v_a^I(t_n)), f \in F_n, n > 0$;
- $v_a^I(P(t_1, t_2, \dots, t_n)) = I|P|(v_a^I(t_1), v_a^I(t_2), \dots, v_a^I(t_n)), P \in P_n, n > 0$;
- $v_a^I(\neg A) = \neg v_a^I(A)$; $v_a^I(A \wedge B) = v_a^I(A) \wedge v_a^I(B)$;
- $v_a^I(A \vee B) = v_a^I(A) \vee v_a^I(B)$; $v_a^I(A \rightarrow B) = v_a^I(A) \rightarrow v_a^I(B)$;
- $v_a^I((\exists x)A(x)) = T$ if and only if $v_{a'}^I(A(x)) = T$ for a function $a' \in [a]_x$
- $v_a^I((\forall x)A(x)) = T$ if and only if $v_{a'}^I(A(x)) = T$ for any function $a' \in [a]_x$

Note: The evaluation of a closed formula U depends only on the interpretation, notation: $v^I(U)$

DEFINITIONS (SEMANTIC CONCEPTS)

- A **formula** A is **satisfiable (consistent)** if there is an interpretation I and an assignment function $a \in As(I)$ such that $v_a^I(A) = T$. Otherwise the formula is **unsatisfiable (inconsistent)**.
- A formula A is **true under the interpretation** I if for any assignment function $a \in As(I)$, $v_a^I(A) = T$, notation: $\models_I A$, and I is called **model** of A .
- A formula A is **false under the interpretation** I if for any assignment function $a \in As(I)$, $v_a^I(A) = F$, and I is called **anti-model** of A .
- A **formula** A is **valid (tautology)** if A is true under all possible interpretations, notation: $\models A$.
- The **formulas** A and B are **logically equivalent** if $v_a^I(A) = v_a^I(B)$ for any interpretation I and any assignment function a , notation: $A \equiv B$.
- A set of formulas S **logically implies** the formula V if all the models of the set S are also models of the formula V . We say that V is a **logical consequence** of the set S , notation: $S \models V$.
- A **set** of formulas is **consistent** if the conjunction of all its formulas has at least one model. A **set** of formulas is **inconsistent** if the conjunction of all its formulas does not have a model.

LOGICAL EQUIVALENCES IN PREDICATE LOGIC

- *Expansion laws*

$$(\forall x)A(x) \equiv (\forall x)A(x) \wedge A(t)$$

– the universal quantifier is an infinitary conjunction,

$$(\exists x)A(x) \equiv (\exists x)A(x) \vee A(t)$$

– the existential quantifier is an infinitary disjunction,
where t is a term which does not contain x .

- *DeMorgan's infinitary laws*

$$\neg(\exists x)A(x) \equiv (\forall x)\neg A(x) \quad \text{and} \quad \neg(\forall x)A(x) \equiv (\exists x)\neg A(x)$$

- *Quantifiers interchanging laws*

$$(\exists x)(\exists y)A(x, y) \equiv (\exists y)(\exists x)A(x, y) \quad \text{and} \quad (\forall x)(\forall y)A(x, y) \equiv (\forall y)(\forall x)A(x, y)$$

Quantifiers of the same type commute!

Note: $(\exists x)(\forall y)B(x, y) \not\equiv (\forall y)(\exists x)B(x, y)$

Quantifiers of different type do not commute!

EXAMPLE 5

Build a **model** and an **anti-model** for the closed predicate formula:

$$U = (\forall x)(P(x) \vee Q(x)) \rightarrow (\forall x)P(x) \vee (\forall x)Q(x).$$

Let us consider the **interpretation** $I_1 = \langle D_1, m \rangle$, where:

$D_1 = \mathbf{N}$ (the set of natural numbers)

$m(P) : \mathbf{N} \rightarrow \{T, F\}, m(P)(x) = "x : 2"$,

$m(Q) : \mathbf{N} \rightarrow \{T, F\}, m(Q)(x) = "x : 3"$.

$$\begin{aligned} v^{I_1}(U) &= v^{I_1}((\forall x)(P(x) \vee Q(x))) \rightarrow v^{I_1}((\forall x)P(x) \vee (\forall x)Q(x)) \\ &= v^{I_1}((\forall x)(P(x) \vee Q(x))) \rightarrow v^{I_1}((\forall x)P(x)) \vee v^{I_1}((\forall x)Q(x)) \\ &= (\forall x)_{x \in \mathbf{N}}(x : 2 \vee x : 3) \rightarrow (\forall x)_{x \in \mathbf{N}}(x : 2) \vee (\forall x)_{x \in \mathbf{N}}(x : 3) \\ &= F \rightarrow F \vee F = F \rightarrow F = T \end{aligned}$$

$v^{I_1}(U) = T$, U is evaluated as true under the interpretation I_1 which is a **model** of U .



EXAMPLE 5 (CONTD.) $U = (\forall x)(P(x) \vee Q(x)) \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$

Let us consider the interpretation $I_2 = \langle D_2, m \rangle$, where:

$D_2 = \{4, 9\}$ – the domain of interpretation;

$m(P) : \{4, 9\} \rightarrow \{T, F\}$, $m(P)(x) = "x:2"$

$m(Q) : \{4, 9\} \rightarrow \{T, F\}$, $m(Q)(x) = "x:3"$.

To evaluate the formula U under the interpretation I_2 , with the finite domain $D_2 = \{4, 9\}$, the universally quantified subformulas are replaced by the conjunction of their instances for $x = 4$ and $x = 9$.

$$\begin{aligned} v^{I_2}(U) &= v^{I_2}((\forall x)(P(x) \vee Q(x))) \rightarrow v^{I_2}((\forall x)P(x) \vee (\forall x)Q(x)) \\ &= v^{I_2}((\forall x)(P(x) \vee Q(x))) \rightarrow v^{I_2}((\forall x)P(x)) \vee v^{I_2}((\forall x)Q(x)) \\ &= (4:2 \vee 4:3) \wedge (9:2 \vee 9:3) \rightarrow (4:2 \wedge 9:2) \vee (4:3 \wedge 9:3) \\ &= (T \vee F) \wedge (F \vee T) \rightarrow (T \wedge F) \vee (F \wedge T) \\ &= T \wedge T \rightarrow F \vee F = T \rightarrow F = F \end{aligned}$$

I_2 evaluates the formula U as false, I_2 is an anti-model of U .



Distributive laws

$(\exists x)(A(x) \vee B(x)) \equiv (\exists x)A(x) \vee (\exists x)B(x)$ distributivity of ' $\exists$ ' over ' $\vee$ '

$(\forall x)(A(x) \wedge B(x)) \equiv (\forall x)A(x) \wedge (\forall x)B(x)$ distributivity of ' $\forall$ ' over ' $\wedge$ '

Semi-distributivity of ' $\exists$ ' over ' $\wedge$ ':

$\models (\exists x)(A(x) \wedge B(x)) \rightarrow (\exists x)A(x) \wedge (\exists x)B(x)$

$\not\models (\exists x)A(x) \wedge (\exists x)B(x) \rightarrow (\exists x)(A(x) \wedge B(x))$

Semi-distributivity of ' $\forall$ ' over ' $\vee$ ':

$\models (\forall x)A(x) \vee (\forall x)B(x) \rightarrow (\forall x)(A(x) \vee B(x))$

$\not\models (\forall x)(A(x) \vee B(x)) \rightarrow (\forall x)A(x) \vee (\forall x)B(x)$

Semi-distributivity of ' $\exists$ ' over ' $\rightarrow$ ':

$\models ((\exists x)A(x) \rightarrow (\exists x)B(x)) \rightarrow (\exists x)(A(x) \rightarrow B(x))$

$\not\models (\exists x)(A(x) \rightarrow B(x)) \rightarrow ((\exists x)A(x) \rightarrow (\exists x)B(x))$

Semi-distributivity of ' $\forall$ ' over ' $\rightarrow$ ':

$\models (\forall x)(A(x) \rightarrow B(x)) \rightarrow ((\forall x)A(x) \rightarrow (\forall x)B(x))$

$\not\models ((\forall x)A(x) \rightarrow (\forall x)B(x)) \rightarrow (\forall x)(A(x) \rightarrow B(x))$

EXAMPLE 6

Prove that the *universal* and *existential* quantifiers do not commute.

We have to prove that the logical equivalence:

$$(\exists x)(\forall y)L(x, y) \equiv (\forall y)(\exists x)L(x, y) \text{ does not hold.}$$

We choose the interpretation $I = \langle D, m \rangle$, where:

- D is the set of all persons in the world
- $m(L) : D \times D \rightarrow \{T, F\}, m(L)(x, y) = "x \text{ loves } y"$

Under the interpretation I , the formula $U_1 = (\exists x)(\forall y)L(x, y)$ has the meaning:

"There exists a person who loves all persons."

The formula $U_2 = (\forall y)(\exists x)L(x, y)$ has the meaning:

"All persons are loved by at least one person.", under the same interpretation I .

These two natural language statements are not equivalent, so $U_1 \neq U_2$,

but note that $U_1 \models U_2$.



PROPERTIES OF FIRST-ORDER LOGIC:

SOUNDNESS, COMPLETENESS, SEMI-DECIDABILITY

Theorem of soundness and completeness states the equivalence between the “logical consequence” concept and the “syntactic consequence” concept.

Let $U_1, \dots, U_{n-1}, U_n, V$ be first-order formulas.

- **completeness:** if $U_1, \dots, U_{n-1}, U_n \models V$ then $U_1, \dots, U_{n-1}, U_n \vdash V$.
- **soundness:** if $U_1, \dots, U_{n-1}, U_n \vdash V$ then $U_1, \dots, U_{n-1}, U_n \models V$.

A particular case of this theorem is the following result:

“A formula is a tautology if and only if it is a theorem in first-order logic.”

Theorem (Church 1936):

The problem of validity of a first-order formula is *undecidable*, but is *semi-decidable*. If a procedure *Proc* is used to check the validity of a first-order formula *U* we have the following cases:

- if *U* is a valid formula, **then** *Proc* ends with the corresponding answer.
- if the formula *U* is not valid, **then** *Proc* ends with the corresponding answer or *Proc* may never stop.